

Enterprises *are* Embracing Open Source PaaS

Platform as a Service is today being used by a large number of enterprises. This article covers the characteristics, adoption patterns and architecture components of a PaaS platform. It also looks at why this service is so popular today.



Platform as a Service (PaaS) is a category of cloud computing services that provides a platform for customers to develop, run and manage applications without the complexity of building and maintaining the infrastructure associated with developing and launching applications. This is the core platform on which cloud native applications and supporting systems are based.

PaaS typically involves diverse application software infrastructure capabilities including application platforms, integration platforms, business analytics platforms, event-streaming services and mobile back-end services. In addition, a PaaS includes a set of monitoring, management, deployment and related capabilities.

There is always a conflict between the developers and the operations teams in the software development process. Developers are keen on getting their environments up and running, without having to wait. The operations team cares about performance and stability. PaaS creates a peaceful environment for both the groups. An application platform delivered as a service is described as PaaS and it is used to

deploy the user code. Cloud Foundry, Cloudify and OpenShift open source environments can be used as PaaS.

Industry adoption of PaaS

451 Research's Market Monitor forecasts that the total market for cloud computing — including PaaS, IaaS and Infrastructure Software as a Service (ITSM, backup, archiving, etc) – will hit US\$ 44.2 billion by 2020.

Gartner predicts that within the PaaS category, the fastest-growing segment is the Database Platform as a Service (dbPaaS), which is expected to reach almost US\$ 10 billion by 2021.

Forbes forecasts that 83 per cent of enterprise workloads will be in the cloud by 2020.

By 2021, enterprise applications will shift towards hyper agile architectures, with 80 per cent of application development being done on cloud platforms (PaaS) using microservices and cloud functions (e.g., AWS Lambda and Azure Functions). Over 95 per cent of new microservices deployed will be in containers (e.g., Docker).